(14) If $A = \begin{bmatrix} \frac{1}{2} & \alpha \\ 0 & \frac{1}{2} \end{bmatrix}$, prove that $\sum_{k=1}^{n} \det(A^k) = \frac{1}{3}\left(1 - \frac{1}{4^n}\right)$.

(15) Without expanding, evaluate the following determinants :

(i) $\begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \\ 6x & 9x & 12x \end{vmatrix}$ (ii) $\begin{vmatrix} x+y & y+z & z+x \\ z & x & y \\ 1 & 1 & 1 \end{vmatrix}$

(16) If A is a square matrix and $|A| = 2$, find the value of $|AA^T|$.

(17) If A and B are square matrices of order 3 such that $|A| = -1$ and $|B| = 3$, find the value of $|3AB|$.

(18) If $\lambda = -2$, determine the value of $\begin{vmatrix} 0 & 2\lambda & 1 \\ \lambda^2 & 0 & 3\lambda^2+1 \\ -1 & 6\lambda-1 & 0 \end{vmatrix}$.

(19) Determine the roots of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$.

(20) Verify that $\det(AB) = (\det A)(\det B)$ for $A = \begin{bmatrix} 4 & 3 & -2 \\ 1 & 0 & 7 \\ 2 & 3 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 3 \\ -2 & 4 & 0 \\ 9 & 7 & 5 \end{bmatrix}$.

(21) Using cofactors of elements of second row, evaluate $|A|$, where $A = \begin{bmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{bmatrix}$.

7.3.3 Application of Factor Theorem to Determinants.

Theorem 7.3 (Factor Theorem)

If each element of a matrix A is a polynomial in x and if $|A|$ vanishes for $x = a$, then $(x - a)$ is a factor of $|A|$.

Note 7.10

(i) This theorem is very much useful when we have to obtain the value of the determinant in 'factors' form.

(ii) If we substitute b for a in the determinant $|A|$, any two of its rows or columns become identical, then $|A| = 0$, and hence by factor theorem $(a - b)$ is a factor of $|A|$.

(iii) If r rows (columns) are identical in a determinant of order n $(n \geq r)$, when we put $x = a$, then $(x - a)^{r-1}$ is a factor of $|A|$.

(iv) A square matrix (or its determinant) is said to be in cyclic symmetric form if each row is obtained from the first row by changing the variables cyclically.

(v) If the determinant is in cyclic symmetric form and if m is the difference between the degree of the product of the factors (obtained by substitution) and the degree of the product of the leading diagonal elements and if

(1) m is zero, then the required factor is a constant k

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